

第十一章作业

教材作业

2、电荷角密度: $\delta = \frac{q}{2\alpha}$

$$\text{对称微元上 } dE = \frac{2\delta d\theta}{4\pi\epsilon_0 r^2} \cdot \cos\theta = \frac{q \cos\theta}{4\pi\epsilon_0 r^2 \alpha} d\theta$$

$$E = \int_0^\alpha \frac{q \cos\theta}{4\pi\epsilon_0 r^2 \alpha} d\theta = \frac{q}{4\pi\epsilon_0 r^2 \alpha} \int_0^\alpha \cos\theta d\theta = \frac{q \sin\alpha}{4\pi\epsilon_0 r^2 \alpha}$$

$$\text{而 } \sin\alpha = \frac{l}{r} = 0.02 \approx 0, \text{ 有 } \frac{\sin\alpha}{\alpha} \approx 1$$

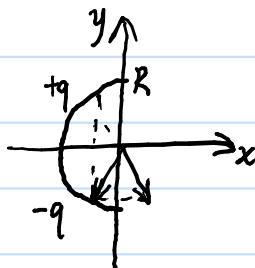
$$\text{代入数值, 有 } E = \frac{q}{4\pi\epsilon_0 r^2} \approx 112.16 \text{ N/C, 方向沿缺口中线向外}$$



4、对称微元上 $dE = \frac{2q \cos\theta}{4\pi\epsilon_0 R^2} dq = \frac{q \cos\theta}{\pi^2 \epsilon_0 R^2} d\theta$

$$E = \frac{q}{\pi^2 \epsilon_0 R^2} \int_0^{\frac{\pi}{2}} \cos\theta d\theta = \frac{q}{\pi^2 \epsilon_0 R^2}$$

$$\text{故 } \vec{E} = -\frac{q}{\pi^2 \epsilon_0 R^2} \vec{j}$$



5、考虑圆环微元

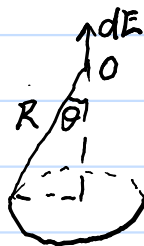
$$dS = 2\pi R \sin\theta \cdot R d\theta = 2\pi R^2 \sin\theta d\theta$$

$$dq = \sigma dS = 2\pi \sigma R^2 \sin\theta d\theta$$

$$dE = \frac{\sigma}{2\epsilon_0} \sin\theta \cos\theta d\theta$$

$$E = \frac{\sigma}{2\epsilon_0} \int_0^{\frac{\pi}{2}} \sin\theta \cos\theta d\theta = \frac{\sigma}{4\epsilon_0}$$

$$\text{故 } \vec{E} = \frac{\sigma}{4\epsilon_0} \vec{j}$$



6、考虑竖直薄片

$$dq = \sigma dS = \sigma_0 \cos\varphi R dp dz$$

$$dE_x = \frac{\cos\varphi dq}{4\pi\epsilon_0 (R^2 + z^2)} \cdot \frac{R}{\sqrt{R^2 + z^2}}, \quad dE_y = \frac{\sin\varphi dq}{4\pi\epsilon_0 (R^2 + z^2)} \cdot \frac{R}{\sqrt{R^2 + z^2}}$$

做二重积分

$$\begin{aligned} E_x &= \int dE_x = \frac{\sigma_0 R^2}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{dz}{(R^2 + z^2)^{\frac{3}{2}}} \int_0^{2\pi} \cos^2\varphi dp \\ &= \frac{\sigma_0 R^2}{4\epsilon_0} \int_{-\infty}^{+\infty} \frac{dz}{(R^2 + z^2)^{\frac{3}{2}}} \\ &= \frac{\sigma_0 R^2}{4\epsilon_0} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\cos\theta}{R^2} \\ &= \frac{\sigma_0}{2\epsilon_0} \end{aligned}$$

$$E_y = \int dE_y = 0$$

$$\text{故 } \vec{E} = \vec{E}_x = -\frac{\sigma}{2\epsilon_0} \vec{i}$$

7. 以 θ 为自变量

$$E = \frac{q}{4\pi\epsilon_0 \left(\frac{d}{\cos\theta}\right)^2} = \frac{q\cos^2\theta}{4\pi\epsilon_0 d^2}$$

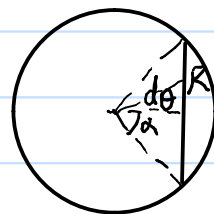
$$\Phi = \int_S \vec{E} \cdot d\vec{S} = \int_S E \cos\theta dS$$

$$= \int_0^\alpha \frac{q\cos^2\theta}{4\pi\epsilon_0 d^2} \cos\theta \cdot 2\pi d^2 \tan\theta \sec^2\theta d\theta$$

$$= \frac{q}{2\epsilon_0} \int_0^\alpha \sin\theta d\theta$$

$$= \frac{q}{2\epsilon_0} (1 - \cos\alpha)$$

其中 $\cos\alpha = \frac{d}{\sqrt{R^2 + d^2}}$, 故 $\Phi = \frac{q}{2\epsilon_0} \left(1 - \frac{d}{\sqrt{R^2 + d^2}}\right)$

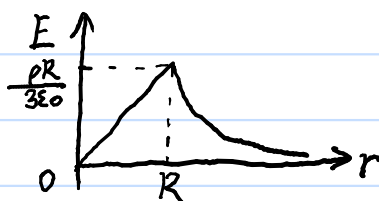


8. 取以 O 为球心, 半径为 r 的高斯面 S

$$\oint_S \vec{E} \cdot d\vec{S} = 4\pi r^2 E = \frac{1}{\epsilon_0} \sum q$$

其中 $q = \begin{cases} \frac{4}{3}\pi r^3 \rho, & r \leq R \\ \frac{4}{3}\pi R^3 \rho, & r > R \end{cases}$

故 $E = \begin{cases} \frac{\rho r}{3\epsilon_0}, & r \leq R \\ \frac{\rho R^3}{3\epsilon_0 r^2}, & r > R \end{cases}$



14. 取半径为 r 的共轴圆筒 S 作为高斯面

$$E \cdot 2\pi r \cdot l = \frac{q}{\epsilon_0}$$

其中 $q = \begin{cases} 0, & r \leq R_1 \\ \lambda l, & R_1 < r \leq R_2 \\ 0, & r > R_2 \end{cases}$, 有 $E = \begin{cases} 0, & r \leq R_1 \\ \frac{\lambda}{2\pi\epsilon_0 r}, & R_1 < r \leq R_2 \\ 0, & r > R_2 \end{cases}$

以无穷远为零电势点, 当 $r > R_2$ 时, $\varphi = 0$

当 $R_1 < r \leq R_2$ 时, $\varphi = \int_r^{R_2} E \cdot dr = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{r}$

当 $r \leq R_1$ 时, $\varphi = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}$

故 $\varphi = \begin{cases} \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}, & r \leq R_1 \\ \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{r}, & R_1 < r \leq R_2 \\ \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}, & r > R_2 \end{cases}$, $U = \varphi(R_1) - \varphi(R_2) = \frac{\lambda}{2\pi\epsilon_0} \ln \frac{R_2}{R_1}$

16. 取半径为 r 的圆筒作为高斯面, 有 $E \cdot 2\pi r \cdot l = \frac{q}{\epsilon_0}$

则 $E = \begin{cases} \frac{\rho r}{2\epsilon_0}, & r \leq R \\ \frac{\rho R^2}{2\epsilon_0 r}, & r > R \end{cases}$

当 $r \leq R$ 时, $\varphi = -\int_0^R E \cdot dr = -\frac{\rho r^2}{4\epsilon_0}$

当 $r > R$ 时, $\varphi = -\frac{\rho R^2}{4\epsilon_0} - \int_R^r E \cdot dr = -\frac{\rho R^2}{4\epsilon_0} (1 + 2\ln \frac{r}{R})$

$$17. dq = \sigma dS = 2\pi\sigma x \sin\theta dx$$

$$\varphi = \int_0^R \frac{dq}{4\pi\epsilon_0 x} = \frac{\sigma}{2\epsilon_0} \int_0^R \sin\theta dx = \frac{\sigma R}{2\epsilon_0}$$

系列化习题

1. 视为完整平面与负电圆孔叠加

$$2E_1 S = \frac{\sigma S}{\epsilon_0}$$

$$E_1 = \frac{\sigma}{2\epsilon_0}$$

$$dE_2 = \frac{dq}{4\pi\epsilon_0(r^2+d^2)} = \frac{-\sigma dr}{2\epsilon_0(r^2+d^2)}$$

$$E_2 = \int_0^R \frac{-\sigma dr}{2\epsilon_0(r^2+d^2)} \cdot \frac{d}{\sqrt{r^2+d^2}} = \frac{-\sigma d}{4\epsilon_0} \int_0^R \frac{dr}{(r^2+d^2)^{3/2}} = \frac{\sigma d}{2\epsilon_0} \left(\frac{1}{\sqrt{R^2+d^2}} - \frac{1}{d} \right)$$

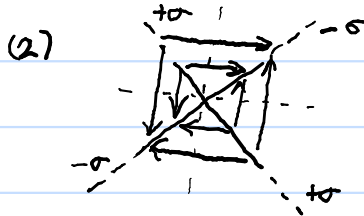
$$\text{故 } E_p = E_1 + E_2 = \frac{\sigma}{2\epsilon_0} \cdot \frac{d}{\sqrt{R^2+d^2}}, \text{ 沿 } OP \text{ 方向}$$

4. 由高斯定理, $2E_1 S = \frac{\sigma S}{\epsilon_0}$

$$E_1 = \frac{\sigma}{2\epsilon_0}$$

$$(1) \vec{E} = \vec{E}_1 + \vec{E}_2$$

故 $E = \frac{\sqrt{2}\sigma}{2\epsilon_0}$, 方向取决于象限, 但总从正极板指向负极板



$$8. (1) \vec{E}_1 = 2ab\vec{i}$$

$$\Phi_1 = E_1 S_1 = 2a^3b$$

$$\Phi_2 = \vec{E}_2 \cdot \vec{S}_2 = 0$$

$$(2) \Phi = \sum_i \Phi_i = a^3b = \frac{q}{\epsilon_0}$$

$$q = \epsilon_0 a^3b$$

11. (1) 取同轴圆筒作为高斯面

$$E \cdot 2\pi r l = \frac{q}{\epsilon_0}$$

$$q = \rho \cdot \pi r'^2 l$$

$$\text{故 } E = \frac{\rho r'}{2\epsilon_0}$$

$$(2) \text{ 视作与负电圆柱叠加, } E_1 = \frac{\rho r}{2\epsilon_0}, E_2 = -\frac{\rho r'}{2\epsilon_0}$$

$$\text{故 } E = E_1 - E_2 = \frac{\rho}{2\epsilon_0}(r+r') = \frac{\rho d}{2\epsilon_0}$$

$$14. (1) U_{ab} = \varphi_a - \varphi_b = (\varphi_a - \varphi_0) - (\varphi_b - \varphi_0) = -E_1 \cdot \frac{a}{2} + E_2 \cdot \frac{a}{2} = \frac{a}{2}(E_2 - E_1)$$

$$(2) \text{由高斯定理, } E_1 \cdot S + E_2 \cdot S = \frac{\sigma S}{\epsilon_0}$$

$$\text{故 } \sigma = \epsilon_0(E_1 + E_2)$$

$$(3) E_0 = \frac{\sigma}{2\epsilon_0} = \frac{E_1 + E_2}{2}$$

$$E' = E_2 - E_0 = \frac{E_2 - E_1}{2}, \text{ 方向水平向右.}$$

补充作业

1. 等效于两完整球体的系统

$$\text{对于单个球体, } E(r) = \begin{cases} \frac{\rho r}{3\epsilon_0}, & r \leq R \\ \frac{\rho R^3}{3\epsilon_0 r^2}, & r > R \end{cases}, \quad \varphi(r) = \begin{cases} \frac{\rho}{6\epsilon_0}(3R^2 + r^2), & r \leq R \\ \frac{\rho R^3}{3\epsilon_0 r}, & r > R \end{cases}$$

$$\varphi_p = -\frac{\rho R^3}{3\epsilon_0 r} + \frac{\rho R^3}{3\epsilon_0(r+d)} = \frac{\rho R^3}{3\epsilon_0} \left(\frac{1}{r+d} - \frac{1}{r} \right)$$

$$\varphi_{01} = \frac{\rho R^2}{2\epsilon_0} - \frac{\rho R^3}{3\epsilon_0 d} = \frac{\rho R^3}{\epsilon_0} \left(\frac{1}{2R} - \frac{1}{3d} \right)$$

$$\varphi_{02} = \frac{\rho R^3}{3\epsilon_0 d} - \frac{\rho R^2}{2\epsilon_0} = \frac{\rho R^3}{\epsilon_0} \left(\frac{1}{3d} - \frac{1}{2R} \right)$$

2. 在板内总有 $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

$$\text{又 } \int_V \nabla \cdot \vec{E} dV = \int_S \vec{E} \cdot d\vec{S}$$

$$\text{取柱面有 } \frac{\rho}{\epsilon_0} \cdot S \cdot 2r = 2ES$$

$$\text{则 } E = \frac{\rho r}{\epsilon_0}, \text{ 其中 } r \text{ 为到板中央平面的距离}$$

$$\text{故 } \vec{E} = \frac{\rho r}{\epsilon_0} \vec{i}, \quad -\frac{D}{2} < r < \frac{D}{2}.$$

3. (1) 加入试探电荷 q_0 , 其电力 $\vec{F} = \frac{q \cdot q_0}{4\pi\epsilon_0} \frac{1-\sqrt{ar}}{r^2} \vec{e}_r$

$$\text{故 } \vec{E} = \frac{\vec{F}}{q_0} = \frac{q}{4\pi\epsilon_0} \frac{1-\sqrt{ar}}{r^2} \vec{e}_r$$

(2) 注意到 $\vec{F} = \vec{F}(\vec{r})$, $F = F(r)$, 可知电力为保守力

$$\text{有 } W = \oint_L q_0 \vec{E} \cdot d\vec{l} = 0$$

$$\text{故 } \oint_L \vec{E} \cdot d\vec{l} = 0$$

$$(3) \oint_S \vec{E} \cdot d\vec{S} = \frac{q}{4\pi\epsilon_0} \cdot \frac{1-\sqrt{ar}}{r_1^2} \cdot 4\pi r_1^2 = \frac{q}{\epsilon_0} (1-\sqrt{ar_1})$$

$$(4) \oint_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0} (1-\sqrt{a(n+dr)})$$

$$\iiint_V \nabla \cdot \vec{E} dV = 4\pi r_1^2 \nabla \cdot \vec{E}_1 \cdot dr = \frac{q}{\epsilon_0} \sqrt{a} (\sqrt{r_1} - \sqrt{r_1+dr}) \approx \frac{q}{2\epsilon_0} \frac{\sqrt{a}}{r_1} dr$$

$$\text{故 } \nabla \cdot \vec{E}(r) = \frac{q\sqrt{a}}{8\pi\epsilon_0} \cdot r_1^{-\frac{5}{2}}$$

与库仑定律相比, 电场强度仍为0, 散度从0变为关于r的函数

附加题

1. 四电荷的电势贡献为

$$\varphi_1 = \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(r\cos\theta + \frac{l}{2})^2 + (r\sin\theta - \frac{l}{2})^2}} \approx \frac{q}{4\pi\epsilon_0 r} \cdot \frac{1}{\sqrt{1 + \frac{l}{r}(\cos\theta - \sin\theta)}}$$

$$\varphi_2 = \frac{-q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(r\cos\theta - \frac{l}{2})^2 + (r\sin\theta + \frac{l}{2})^2}} \approx \frac{-q}{4\pi\epsilon_0 r} \cdot \frac{1}{\sqrt{1 + \frac{l}{r}(-\cos\theta - \sin\theta)}}$$

$$\varphi_3 = \frac{-q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(r\cos\theta + \frac{l}{2})^2 + (r\sin\theta + \frac{l}{2})^2}} \approx \frac{-q}{4\pi\epsilon_0 r} \cdot \frac{1}{\sqrt{1 + \frac{l}{r}(\cos\theta + \sin\theta)}}$$

$$\varphi_4 = \frac{q}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{(r\cos\theta - \frac{l}{2})^2 + (r\sin\theta - \frac{l}{2})^2}} \approx \frac{q}{4\pi\epsilon_0 r} \cdot \frac{1}{\sqrt{1 + \frac{l}{r}(-\cos\theta + \sin\theta)}}$$

当 $x \rightarrow 0$ 时, 有 $(1+x)^{-\frac{1}{2}} = 1 - \frac{1}{2}x + \frac{3}{8}x^2 + o(x^2)$, 于是

$$\varphi_1 \approx \frac{q}{4\pi\epsilon_0 r} \left[1 - \frac{1}{2} \frac{l}{r} (\cos\theta - \sin\theta) + \frac{3}{8} \frac{l^2}{r^2} (\cos\theta - \sin\theta)^2 \right]$$

$$\varphi_2 \approx \frac{-q}{4\pi\epsilon_0 r} \left[1 - \frac{1}{2} \frac{l}{r} (-\cos\theta - \sin\theta) + \frac{3}{8} \frac{l^2}{r^2} (-\cos\theta - \sin\theta)^2 \right]$$

$$\varphi_3 \approx \frac{-q}{4\pi\epsilon_0 r} \left[1 - \frac{1}{2} \frac{l}{r} (\cos\theta + \sin\theta) + \frac{3}{8} \frac{l^2}{r^2} (\cos\theta + \sin\theta)^2 \right]$$

$$\varphi_4 \approx \frac{q}{4\pi\epsilon_0 r} \left[1 - \frac{1}{2} \frac{l}{r} (-\cos\theta + \sin\theta) + \frac{3}{8} \frac{l^2}{r^2} (-\cos\theta + \sin\theta)^2 \right]$$

故 $\varphi = (\varphi_1 + \varphi_4) + (\varphi_2 + \varphi_3)$

$$= \frac{q}{4\pi\epsilon_0 r} \left[2 + \frac{3}{4} \cdot \frac{l^2}{r^2} (\cos\theta - \sin\theta)^2 \right] - \frac{q}{4\pi\epsilon_0 r} \left[2 + \frac{3}{4} \cdot \frac{l^2}{r^2} (\cos\theta + \sin\theta)^2 \right]$$

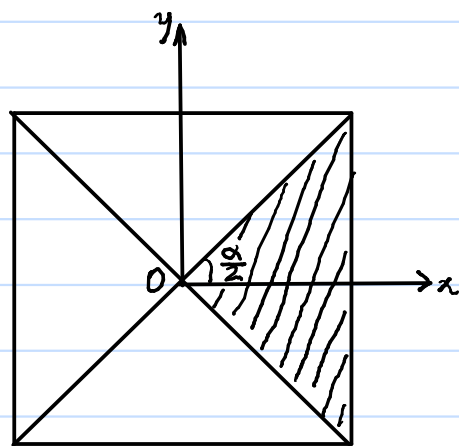
$$= \frac{q}{4\pi\epsilon_0 r} \cdot \frac{3}{4} \cdot \frac{l^2}{r^2} \cdot (-4\sin\theta\cos\theta)$$

$$= \frac{-3ql^2\sin\theta\cos\theta}{4\pi\epsilon_0 r^3} \quad (r \gg l)$$

2、将 N 边形划分为 N 个三角形，考察其中一块的贡献

$$\text{顶角 } \alpha = \frac{2\pi}{N}, \quad \text{半角 } \frac{\alpha}{2} = \frac{\pi}{N}$$

$$\begin{aligned} \text{电势贡献 } \varphi_1 &= \iint_S \frac{\sigma dS}{4\pi\epsilon_0 r} \\ &= \int_0^a dx \int_{-x\tan\frac{\alpha}{2}}^{x\tan\frac{\alpha}{2}} \frac{\sigma}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{x^2+y^2}} dy \\ &= \frac{\sigma}{4\pi\epsilon_0} \int_0^a \ln \frac{1+\sin\frac{\alpha}{2}}{1-\sin\frac{\alpha}{2}} dx \\ &= \frac{\sigma a}{4\pi\epsilon_0} \ln \frac{1+\sin\frac{\pi}{N}}{1-\sin\frac{\pi}{N}} \end{aligned}$$



$$\text{故 } \varphi_0 = N\varphi_1 = \frac{\sigma a}{4\pi\epsilon_0} N \cdot \ln \frac{1+\sin\frac{\pi}{N}}{1-\sin\frac{\pi}{N}}$$

令 $N \rightarrow \infty$ 有

$$\lim_{n \rightarrow \infty} n \cdot \ln \frac{1+\sin\frac{\pi}{n}}{1-\sin\frac{\pi}{n}} = \lim_{x \rightarrow 0} \pi \frac{\ln(1+\sin x) - \ln(1-\sin x)}{x} = 2\pi$$

$$\text{故 } \varphi_0|_{N \rightarrow \infty} = \frac{\sigma a}{2\epsilon_0}, \quad \text{与圆面电势公式一致}$$